



BAHIRDAR UNIVERSITY
INSTITUTE OF TECHNOLOGY (BIT)
FACULTY OF COMPUTING
DEPARTMENT OF INFORMATION SYSTEM

Submitted By Hanamariam Biruk

ID NO 1601697

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1 Introduction (Background and Motivation)

1.1 Background

An Operating System (OS) is the fundamental software that manages computer hardware and software resources, providing essential services for computer programs. In modern computing, virtualization technology enables users to run multiple OS instances simultaneously on a single physical machine, creating isolated and secure environments for testing, development, and learning.

Windows 8 is a Microsoft operating system released in 2012, notable for introducing a touch-friendly interface, improved performance, and enhanced integration with cloud services. Though succeeded by newer versions, Windows 8 remains a relevant case study in OS evolution, interface design, and system compatibility.

Using Oracle VM VirtualBox, Windows 8 can be installed and evaluated in a virtual environment without affecting the host system. This approach provides a practical, risk-free way to explore OS installation processes, driver management, system configuration, and user interface interaction.

1.2 Motivation

The motivation for this project is to:

- Gain hands-on experience with installing and configuring a Microsoft Windows operating system in a virtualized environment.
 - Understand the architectural and interface changes introduced in Windows 8 compared to earlier versions.
 - Explore virtualization concepts and learn how to configure virtual hardware for optimal performance.
 - Develop troubleshooting and system setup skills applicable to real-world IT and support scenarios.
 - Compare and contrast Windows and Linux installation processes, system tools, and user environments.
- Here are the Objectives of the Project, structured to align with the format used

in your openSUSE PDF.

2 Objectives of the Project

The main objectives of this project work are:

1. To demonstrate the Windows 8 installation process in a virtualized environment using Oracle VM VirtualBox.
2. To configure a virtual machine with appropriate hardware settings (RAM, storage, networking) to support Windows 8.
3. To understand the Windows 8 boot process, partitioning scheme, and filesystem (NTFS) configuration during installation.
4. To explore the Windows 8 user interface and features, including the Start Screen, desktop environment, and built-in applications.
5. To gain practical experience with post-installation tasks such as driver installation, Windows Update, and basic system configuration.
6. To compare Windows and Linux installation approaches, noting differences in partitioning, user setup, and system tools.
7. To identify and resolve common installation challenges in a virtual environment, such as boot issues, driver compatibility, and activation.
8. To prepare foundational skills for managing Windows-based systems in academic, lab, or professional IT environments.

Here is the Requirements section for your Windows 8 installation project, written in a clean, structured format without references, as you requested. References to books and resources will be compiled separately at the end of the project.

3 Requirements

This section outlines the hardware and software requirements necessary to install and run Windows 8 in a virtualized environment using Oracle VM VirtualBox.

3.1 Hardware Requirements

Minimum Hardware Requirements (for the host machine)

- Processor (CPU): 64-bit processor with hardware virtualization support (Intel VT-x or AMD-V)
- RAM (Main Memory): 4 GB minimum
- Storage: At least 20 GB free disk space for the virtual machine
- Display: 1024×768 screen resolution or higher
- Input Devices: Keyboard and mouse

Recommended Hardware Requirements

- Processor: Dual-core or better 64-bit CPU with virtualization enabled in BIOS/UEFI
- RAM: 8 GB or more for smooth host and guest performance
- Storage: 40 GB or more free space (for VM disk, snapshots, and system updates)
- Internet Connection: Required for Windows activation, updates, and driver downloads

3.2 Software Requirements

- Host Operating System: Windows 10/11, macOS (Intel-based), or a modern Linux distribution
- Virtualization Software: Oracle VM VirtualBox (version 7.0 or later recommended)
- Guest OS Image: Windows 8 ISO (x64 version recommended for compatibility)
- Optional: VirtualBox Extension Pack for enhanced USB, RDP, and encryption support

Here is a clear, step-by-step explanation of the Windows 8 installation process within Oracle VM VirtualBox, suitable for your project documentation.

4 Installation of Windows 8 (VirtualBox)

4.1 Creating and Configuring the Virtual Machine

1. Open VirtualBox and click New.
2. Enter the VM name: `Windows 8 VM`.
3. Select Type: Microsoft Windows.
4. Select Version: Windows 8 (64-bit).
5. Allocate RAM: 4096 MB (4 GB).

6. Create a virtual hard disk: 40 GB, VDI format, dynamically allocated.
7. Click Create to finalize the VM.

4.2 Attaching the Windows 8 ISO

1. Select the VM and click Settings.
2. Go to Storage.
3. Under Controller: IDE, click the empty optical drive icon.
4. Choose Disk Image and browse to select the Windows 8 ISO file.
5. Click OK to save changes.

4.3 Starting the Installation

1. Start the VM. It will boot from the ISO.
2. The Windows Setup screen will appear.
3. Select your Language, Time format, and Keyboard layout.
4. Click Next and then Install now.

4.4 Entering the Product Key

1. Enter your Windows product key when prompted.
2. Alternatively, you can skip this step and activate later.
3. Click Next.

4.5 Accepting the License Terms

1. Read the Microsoft Software License Terms.
2. Check the box: I accept the license terms.
3. Click Next.

4.6 Choosing the Installation Type

1. Select Custom: Install Windows only (advanced).
2. This option allows full control over disk partitioning.

4.7 Disk Partitioning

1. You will see the virtual disk listed (40 GB unallocated space).
2. Select the disk and click New.
3. Specify the partition size (use the full disk for simplicity).

4. Click Apply. Windows will create necessary system partitions.
5. Select the primary partition and click Next to begin installation.

4.8 Installing Windows Files

1. The system will copy files, install features and updates, and complete installation.
2. This process may take 15-30 minutes.
3. The VM will restart automatically once or twice during installation.

4.9 Personalizing Windows

1. After restart, you will see the Personalize screen.
2. Choose a PC name (e.g., `Win8-VM`).
3. Click Next.

4.10 Settings Configuration

1. You can choose Use express settings for default configurations.
2. Alternatively, select Customize to adjust privacy and update settings manually.
3. Click Next to proceed.

4.11 Signing In

1. You can sign in with a Microsoft account or create a local account.
2. For simplicity, choose Create a local account.
3. Enter a username and password (optional).
4. Click Finish.

4.12 Final Setup

1. Windows will finalize settings and prepare the desktop.
2. You will be greeted with the Windows 8 Start Screen.
3. Installation is now complete.

5 ISSUES

1. Incompatibility with Modern Hardware
2. Windows 8 lacks native drivers for newer hardware components such as USB 3.1/Type-C ports,

NVMe SSDs, and modern Wi-Fi/Bluetooth adapters. This can lead to installation failures, unrecognized storage devices, or limited functionality post-installation.

2. Secure Boot and UEFI Boot Issues

During installation, systems with UEFI firmware and Secure Boot enabled often encountered boot errors or installation failure. This was especially problematic for users attempting to dual-boot with other operating systems or install on older hardware with legacy BIOS.

3. User Interface Disorientation

The introduction of the touch-optimized Start Screen confused traditional desktop users accustomed to the Start Menu. Many users struggled with navigation, finding settings, and accessing system tools, leading to a steep learning curve and reduced productivity after installation.

4. Activation and Update Failures

Windows 8 activation servers are no longer actively supported, making product key validation difficult or impossible. Additionally, Windows Update services for Windows 8 have been discontinued, leaving the system vulnerable and unable to receive critical security patches or driver updates post-installation.

6 Solutions for Windows 8 Installation Challenges

1. Use Virtualization for Safe Installation

– Solution: Install Windows 8 in a virtual machine (e.g., VirtualBox, VMware) instead of on physical hardware.

– Benefit: Isolates the outdated OS from your main system, avoids driver issues, and allows safe experimentation without security risks to the host.

2. Install Windows 8.1 Instead

– Solution: Upgrade to Windows 8.1 (a free update from Windows 8) which has extended support until January 2024 and includes important UI improvements and updates.

– Benefit: Better compatibility, slightly more recent security patches, and the return of the Start button.

3. Use Offline Installation & Local Drivers

– Solution: Prepare necessary drivers (network, storage, graphics) on a USB drive before

installation. Use offline installation media to avoid update and activation failures during setup.

- Benefit: Reduces installation failures due to missing drivers and avoids problematic online activation during setup.

4. Implement Strong Network Isolation

- Solution: If Windows 8 must be used, keep it offline or behind a firewall. Use it only in a isolated network or air-gapped environment.

- Benefit: Minimizes exposure to security vulnerabilities and malware that target the unsupported OS.

5. Dual-Boot or Use as a Legacy System Only

- Solution: Install Windows 8 in a dual-boot configuration for limited, specific use—such as running legacy software—and use a modern OS (Windows 10/11 or Linux) for daily tasks.

- Benefit: All access to old applications while maintaining security and productivity through a modern operating system.

6. Use Windows 8 Only for Academic/Lab Purposes

- Solution: Clearly scope the use of Windows 8 to virtual labs, coursework, or OS studies only. Do not use it for personal data, online banking, or any sensitive activities.

- Benefit: Meets learning objectives without compromising security or functionality in real-world use.

7 Why, How, and When Windows 8 Became Outdated

7.1 When Windows 8 Became Outdated: The Timeline of Obsolescence

Windows 8 entered its phase of obsolescence through a structured and publicly communicated timeline established by Microsoft's software lifecycle policy. The initial release in 2012 was followed by a significant update, Windows 8.1, in 2013, which was treated as a separate servicing entity. Microsoft concluded mainstream support for Windows 8 on January 9, 2018. This milestone marked the end of free incident support, warranty claims, design changes, and non-security updates. The more critical endpoint arrived on January 10, 2023, when Microsoft terminated extended support. The conclusion of extended support is the definitive moment an operating system becomes fully outdated in the enterprise and security context, as it ceases to receive critical security updates

and technical assistance. This lifecycle model, common to commercial software, ensures products have a predictable shelf life, after which they are considered legacy and pose operational risks if deployed in modern environments.

7.2 How Windows 8 Became Outdated: The Technical and Ecosystem Evolution

The obsolescence of Windows 8 was not merely a date on a calendar but a process driven by rapid advancements in hardware, software, and security paradigms. Technologically, the computing landscape evolved significantly post-2012. New hardware interfaces like USB-C, NVMe storage, and advanced power management required kernel-level support and new drivers, which were not developed for the Windows 8 driver model. Concurrently, software development shifted towards continuous integration and cloud-based services that increasingly relied on APIs and security protocols built into newer Windows versions. The industry-wide migration to UEFI Secure Boot and TPM-based security also left Windows 8 behind, as its implementation was an early version that lacked refinements and broader compatibility. Furthermore, the rise of sophisticated cyber threats necessitated ongoing security research and patching, a process that halted for Windows 8 after its support ended, leaving known vulnerabilities permanently unaddressed and creating an ever-widening gap between its capabilities and modern security requirements.

7.3 Why Windows 8 Became Outdated: Strategic, Market, and Architectural Drivers

The reasons for Windows 8's obsolescence are rooted in strategic business decisions, market response, and fundamental architectural shifts. Strategically, Microsoft operates on a "Modern Lifecycle Policy" for its flagship products, deliberately phasing out older versions to consolidate development resources, streamline support channels, and incentivize adoption of newer platforms with updated revenue models, such as cloud services integration in Windows 10 and 11. The market itself rejected key aspects of Windows 8; its radical touch-centric interface was poorly received by the dominant desktop user base, leading to public criticism and slow adoption. This tepid response accelerated Microsoft's push toward its successor. Architecturally, Windows 8 represented a transitional hybrid between the classic Windows desktop and a touch-first, app-centric model. This incomplete vision was superseded by the more cohesive and service-oriented architectures of Windows 10 and 11, which offered a unified update mechanism, a refined hybrid

interface, and a platform built for ongoing “Windows-as-a-Service” delivery. Thus, Windows 8 was outdated not only by time but by its inability to fulfill a sustainable, long-term role in Microsoft’s evolving ecosystem and the broader market’s needs.

8 File System Support in Windows 8

8.1 NTFS: The Primary File System

Windows 8 maintained NTFS as its default and primary file system, inheriting its robustness, security, and reliability from previous Windows versions. NTFS in Windows 8 provided advanced features including file-level encryption through Encrypting File System, disk quotas, volume shadow copy for system backups, and support for large volume and file sizes—up to 256 TB per volume and 16 TB per individual file. It also supported access control lists for detailed permission management, which was essential for multi-user environments and enterprise deployments. The file system’s journaling capability ensured greater data integrity by logging changes before they were committed, reducing the risk of corruption during unexpected system shutdowns.

8.2 FAT32 and exFAT: Legacy and Removable Media Support

For compatibility with removable storage devices and older systems, Windows 8 continued to support FAT32 and exFAT file systems. FAT32 was primarily used for USB flash drives, memory cards, and other external storage devices that required interoperability with non-Windows operating systems or older electronic devices. However, due to its limitations—such as a maximum file size of 4 GB—it was not suitable for modern storage needs. exFAT was introduced as a successor to FAT32, offering support for larger files and volumes while maintaining compatibility across platforms. Windows 8 utilized exFAT for high-capacity removable media like external hard drives and SDXC cards, providing a balance between performance and cross-platform usability without the file size constraints of FAT32.

8.3 ReFS: Introduction of the Resilient File System

Windows 8 marked the introduction of ReFS (Resilient File System), though its implementation was initially limited to Windows Server 2012. ReFS was designed to address the increasing demands of data storage in terms of scalability, reliability, and automatic error correction. It featured built-in resilience to data corruption through integrity streaming and automated repair

mechanisms. While not available as a bootable file system for client versions of Windows 8, its inclusion in the server edition signaled Microsoft's direction toward more robust storage solutions for handling large datasets and critical applications, laying the groundwork for its expanded role in later Windows versions and enterprise storage environments.

8.4 Lack of Native Linux File System Support

Unlike some contemporary and subsequent operating systems, Windows 8 did not offer native read/write support for Linux-based file systems such as ext4 or Btrfs. Users requiring access to these file systems needed to rely on third-party software solutions, which often provided limited functionality and potential stability risks. This lack of native support highlighted the operating system's focus on its proprietary ecosystem and primary use cases, rather than cross-platform file system interoperability. It reinforced the separation between Windows and Unix-like systems in terms of storage management, requiring additional tools or configurations for users operating in mixed-environment settings.

8.5 Storage Spaces and Virtual Disk Support

Windows 8 introduced Storage Spaces, a storage virtualization feature that allowed users to pool physical disks into logical volumes with optional resiliency through mirroring or parity. While not a file system itself, Storage Spaces operated in conjunction with NTFS or ReFS to provide flexible software-defined storage management. This feature enabled users to create large, fault-tolerant storage pools without requiring specialized hardware. Additionally, Windows 8 supported mounting and managing virtual hard disk files (VHD and VHDX formats) natively, allowing them to be treated as physical disks for file storage and system imaging. This integration enhanced the operating system's utility in virtualized and development environments, bridging physical and virtual storage management.

8.6 Why These File System Choices Were Made in Windows 8

The file system architecture of Windows 8 was not arbitrary; it was the result of deliberate engineering and strategic decisions aimed at balancing **backward compatibility, performance, security, and future readiness**. At its core, the selection and implementation of file systems were driven by the need to maintain a stable platform for both consumer and enterprise users while cautiously introducing new technologies without disrupting existing workflows. NTFS remained the

default because it was a mature, trusted, and feature-rich system capable of meeting the security and reliability demands of modern computing—especially in business environments where encryption, permissions, and auditing are non-negotiable. Retaining FAT32 and exFAT support was essential for universal interoperability with removable media, cameras, gaming consoles, and other embedded systems, ensuring Windows 8 devices could interact seamlessly with a wide array of external storage without requiring user reformatting.

The introduction of ReFS, even in a limited server-only capacity, reflected Microsoft's forward-looking approach to data integrity at scale—anticipating the rise of big data, virtualization, and storage-heavy applications. However, its exclusion from the client version of Windows 8 was a calculated risk management decision; ReFS was still considered nascent and unsuitable for boot volumes or general consumer use. The omission of native Linux file system support underscored Microsoft's continued focus on its own ecosystem, prioritizing performance and security integration over cross-platform convenience—a stance that also mitigated potential support complexities and intellectual property considerations. Finally, features like Storage Spaces and native VHD support were included to empower users and IT administrators with software-defined storage flexibility, aligning with industry trends toward virtualization and pooled storage resources without relying on expensive hardware solutions. In summary, Windows 8's file system support was designed to be pragmatically comprehensive: rooted in proven technology, cautiously innovative where needed, and meticulously aligned with both current user expectations and emerging IT paradigms.

9 Advantages and Disadvantages of Windows 8

9.1 Advantages of Windows 8

Windows 8 introduced several forward-thinking features and improvements that distinguished it from its predecessors and influenced later versions of Windows. One of its most significant advantages was its **touch-optimized interface**, which included the Live Tile-based Start Screen designed to bridge the gap between traditional desktops and emerging touch-screen devices like tablets and hybrids. This made Windows 8 one of the first truly cross-platform operating systems from Microsoft, capable of running on both desktop PCs and mobile hardware with a consistent user experience. The improved boot time was another notable benefit, thanks to a hybrid boot system

that combined aspects of cold booting and hibernation, allowing the system to start significantly faster than Windows 7. Additionally, Windows 8 featured **enhanced security measures, including Secure Boot to prevent rootkit attacks, improved Windows Defender with real-time protection, and stricter app vetting through the Windows Store. The operating system also offered better resource management and performance efficiency, with reduced memory footprint and improved multi-monitor support, making it suitable for both low-end hardware and productivity-focused multi-screen setups. Integration with Microsoft cloud services like OneDrive and Microsoft Account synchronization provided seamless access to files and settings across devices, reflecting a shift toward connected, service-based computing.

9.2 Disadvantages of Windows 8

Despite its innovations, Windows 8 faced considerable criticism and practical shortcomings that limited its adoption and long-term viability. The most prominent disadvantage was its radical and confusing user interface, which replaced the familiar Start Menu with a full-screen Start Screen optimized for touch but disruptive for traditional mouse-and-keyboard users. This led to a steep learning curve, user frustration, and a sense of disorientation that alienated many longtime Windows users. The split between two disparate environments**the modern “Metro” app interface and the classic Windows desktopcreated a fragmented experience where apps could not run in windows, manage files easily, or integrate consistently with desktop workflows. Furthermore, limited app availability in the Windows Store at launch hampered the ecosystem’s growth, especially compared to iOS and Android, leaving users with few compelling touch-optimized applications. From a technical standpoint, while Windows 8 introduced new features, it also lacked polish in key areas; for example, certain settings were buried in the new Metro-style Control Panel while others remained in the classic version, creating configuration confusion. Lastly, the early termination of mainstream support and the subsequent rapid shift to Windows 8.1 and Windows 10 signaled Microsoft’s own recognition of the system’s shortcomings, leaving Windows 8 as a short-lived, transitional OS that failed to gain lasting traction in the consumer or enterprise markets.

10 Conclusion

Windows 8 serves as a compelling study in operating system evolution—a system defined by visionary

ambition and tangible missteps. Its introduction of a touch-first interface represented a bold departure from decades of desktop convention, seeking to unify mobile and traditional computing experiences. This radical redesign, while forward-thinking, disrupted established user workflows and introduced a dual-environment experience that often felt fragmented rather than unified. The system's technical merits were notable, with measurable improvements in boot performance, security architecture, and resource efficiency. Its early embrace of cloud integration and modern hardware support reflected a clear intent to push the Windows platform toward a more connected, responsive future. Yet, these advancements were overshadowed by fundamental usability challenges that limited adoption and ultimately defined its legacy as a transitional release.

Windows 8's relatively brief lifecycle and rapid evolution into Windows 8.1—and later Windows 10—demonstrate how quickly an operating system can become outdated when its core interaction model fails to align with user expectations. Today, its discontinuation and lack of security updates render it unsuitable for practical use, yet it remains a valuable case study in the risks and rewards of ambitious platform transformation.

In the context of operating system history, Windows 8 stands as a reminder that innovation must be balanced with intuition, and that even the most technically sound systems cannot succeed without genuine user acceptance. Its legacy continues to inform modern OS design, serving as both an inspiration for integration and a cautionary tale of disruption without adequate continuity.

11 Future Outlook & Recommendations

11.1 Future Outlook

Windows 8 is firmly positioned as a legacy operating system with no future in mainstream or enterprise computing. Its lifecycle has concluded, and it will not receive security updates, compatibility patches, or official support moving forward. As technology continues to advance rapidly, Windows 8 will become increasingly isolated—unable to run modern software, incompatible with emerging hardware, and exposed to unmitigated security threats. It will likely remain useful only in highly specialized, isolated, or educational contexts, such as digital preservation projects, retro computing, and academic studies of operating system evolution. In the broader technology ecosystem, Windows 8 will serve primarily as a historical reference point—a case study in interface design, platform transitions, and the challenges of unifying touch and desktop

paradigms.

11.2 Recommendations

1. For Educational & Academic Use

- Recommendation: Continue using Windows 8 in virtualized environments for teaching operating system concepts, installation procedures, and UI/UX comparison studies.
- Why: It provides a safe, controlled setting to explore a significant but discontinued OS without security risks or hardware limitations.

2. For Legacy System Maintenance

- Recommendation: If Windows 8 must remain in use for legacy software or hardware, air-gap the system from networks and implement strict access controls.
- Why: Isolating the system mitigates security vulnerabilities while allowing continued operation of dependent legacy applications.

3. For General Users & Organizations

- Recommendation: Migrate to a supported operating system such as Windows 10, Windows 11, or a modern Linux distribution.
- Why: Ensures ongoing security updates, software compatibility, hardware support, and access to contemporary features and support.

4. For Developers & IT Professionals

- Recommendation: Study Windows 8 as a transitional artifact in OS design, but develop and test on current platforms.
- Why: Understanding past design decisions can inform future development, but practical work requires modern, secure, and supported environments.

5. For Future OS Design & Research

- Recommendation: Analyze Windows 8's interface and integration challenges as a reference for designing adaptive, user-centered systems.
- Why: Its successes and failures offer valuable lessons in balancing innovation with usability—a continuing challenge in human-computer interaction.

12 Virtualization in Modern Operating Systems

12.1 What is Virtualization?

Virtualization is a technology that allows multiple simulated computing environments to run on a single physical machine. By using a software layer called a hypervisor, physical hardware resources—such as the CPU, memory, storage, and network—are abstracted and partitioned into isolated virtual machines (VMs). Each VM operates as an independent system, capable of running its own operating system and applications, without direct awareness that it shares physical resources.

12.2 Why is Virtualization Used?

Virtualization is widely adopted due to several key advantages:

- Resource Efficiency: Multiple VMs can run on one physical server, optimizing hardware usage and reducing costs.
- Isolation and Security: Each VM is isolated from others, preventing failures or security breaches in one VM from affecting others.
- Testing and Development: Developers can create disposable, reproducible environments to test software without risk to the host system.
- Legacy System Support: Older operating systems and software can run on modern hardware.
- Disaster Recovery and Portability: VMs can be easily backed up, migrated, and restored, improving business continuity and flexibility.

12.3 How Does Virtualization Work?

Virtualization is implemented through the following methods and components:

- Hypervisor Types:
 - Type 1 (Bare-Metal): Runs directly on the host hardware (e.g., VMware ESXi, Microsoft Hyper-V).
 - Type 2 (Hosted): Runs as an application on an existing operating system (e.g., Oracle VirtualBox, VMware Workstation).
- Hardware Assistance: Modern processors provide virtualization extensions (Intel VT-x, AMD-V) to improve performance and security.
- Resource Virtualization:

- CPU time-slicing and memory mapping allow multiple VMs to share physical resources.
- I/O virtualization uses emulated or direct-access hardware to manage storage and network devices.
- Containerization: A lightweight alternative that shares the host OS kernel while isolating user-space processes (e.g., Docker, Kubernetes).

12.4 Real-World Applications

- Cloud Computing: The foundation of Infrastructure-as-a-Service (IaaS) platforms like AWS, Azure, and Google Cloud.
- Data Center Consolidation: Reduces the number of physical servers needed, lowering energy and space costs.
- Desktop Virtualization: Enables remote access to virtual desktops from various devices.
- Software Development and CI/CD: Provides consistent environments for building, testing, and deploying applications.

12.5 Challenges and Considerations

While powerful, virtualization introduces certain challenges:

- Performance Overhead: Virtualization layers can introduce latency, especially for I/O-intensive workloads.
- Resource Contention: Overcommitting resources can degrade performance across VMs.
- Security Risks: Vulnerabilities in the hypervisor can compromise all hosted VMs.
- Management Complexity: Requires specialized tools for provisioning, monitoring, and maintaining virtual environments.

12.6 Future of Virtualization

Virtualization continues to evolve with:

- Nested Virtualization: Running VMs inside other VMs, useful for testing and development.
- Edge Computing: Lightweight virtualization enabling distributed computing closer to data sources.
- Convergence with Containers: Tools like Kubernetes managing both VMs and containers in unified platforms.
- Hardware Innovations: Continued CPU, memory, and network enhancements to reduce virtualization.

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